

Shuigen Liu

10 Lower Kent Ridge Road
Block S17, Room 07-25, Singapore 119076
Email: shuigen.liu@nus.edu.sg
Homepage: <https://liu-shuigen.github.io>

Employment

2025 – present **Research Fellow**, Department of Mathematics, NUS

Education

2021 – 2025 **Ph.D. in Mathematics**, National University of Singapore
Advisors: Prof. Xin T. Tong and Prof. Weizhu Bao
Visiting student, MIT, Jul–Dec 2024 (Host: Prof. Youssef Marzouk)

2017 – 2021 **B.S. in Mathematics**, Peking University

Research Interests

My research lies at the intersection of computational mathematics, applied probability and statistics. I focus on analysis and development of dimension reduction methods for sampling and inverse problems, with applications in Bayesian inference and uncertainty quantification. I also work on ensemble Kalman methods for inverse problems and data assimilation.

Keywords: Sampling, uncertainty quantification, localization, dimension reduction, data assimilation.

Publications

- [5] Localized diffusion models. Georg A. Gottwald, **Shuigen Liu**, Youssef Marzouk, Sebastian Reich, and Xin T. Tong, arXiv:2505.04417 (2025).
- [4] Stein’s method for marginals on large graphical models. Tiangan Cui, **Shuigen Liu**, and Xin T. Tong, arXiv:2410.11771 (2025).
- [3] Estimate of Koopman modes and eigenvalues with Kalman filter. Ningxin Liu, **Shuigen Liu**, Xin T. Tong, and Lijian Jiang, arXiv:2410.02815 (2024).
- [2] Local MALA-within-Gibbs for Bayesian image deblurring with total variation prior. Rafael Flock, **Shuigen Liu**, Yiqiu Dong, and Xin T. Tong, *SIAM J. Sci. Comput.*, 47 (2025), pp. A2127–A2153.
- [1] Dropout ensemble Kalman inversion for high dimensional inverse problems. **Shuigen Liu**, Sebastian Reich and Xin T. Tong, *SIAM J. Numer. Anal.*, 63 (2025), pp. 685–715.

Teaching

National University of Singapore

TA Finite Element Method, Spring 2025
TA Essential Data Analytics Tools: Numerical Computation, Spring 2024
TA Calculus for Computing, Spring 2023 and Fall 2023

Honors and Awards

2024 Faculty of Science PhD Conference Award, National University of Singapore
2024 Overseas Research Immersion Award, National University of Singapore
2021 President’s Graduate Fellowship, National University of Singapore

2021 Outstanding Undergraduate, Peking University

Service

Reviewer for: SIAM/ASA Journal on Uncertainty Quantification, Journal of Machine Learning Research, Statistics and Computing.

Conferences

- | | |
|-----------------|---|
| Jul 2025 | Efficient Sampling Algorithms for Complex Models, IMS NUS, Singapore. |
| Jun 2025 | MATRIX Research Program: Bayesian Learning of Very High-Dimensional Physical Process Models, MATRIX, Melbourne, Australia. |
| May 2025 | 2025 STU-NUS PhD Forum, Shanghai Jiao Tong University, Shanghai, China. |
| Oct 2024 | FDS Conference: Recent Advances and Future Directions for Sampling, Yale University, New Haven, U.S. |
| Feb 2024 | 2024 SIAM Conference on Uncertainty Quantification (UQ24), Trieste, Italy. |
| Aug 2023 | 2023 International Congress on Industrial and Applied Mathematics (ICIAM 2023), Waseda University, Tokyo, Japan. |
| Jun 2023 | Foundations of Computational Mathematics 2023 Conference (FoCM 2023), Sorbonne University, Paris, France. |
| Jul 2022 | 2022 International Conference on Scientific Computation and Differential Equations (SciCADE 2022), University of Iceland, Reykjavík, Iceland. |